

n -SCRAMBLED CANTOR SETS

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ABSTRACT. For all $n \geq 2$, we produce a homeomorphism of $2^{\mathbb{N}}$ with an n -scrambled Cantor set containing an $(n + 1)$ -scrambled uncountable set, but with no $(n + 1)$ -scrambled Cantor set.

INTRODUCTION

Given a metric space X , $f: X \rightarrow X$, and $n \geq 2$, a sequence $x \in X^n$ is *proximal* if $\liminf_{i \rightarrow \infty} \max_{j < k < n} d_X(f^i(x(j)), f^i(x(k))) = 0$, *separated* if $\limsup_{i \rightarrow \infty} \min_{j < k < n} d_X(f^i(x(j)), f^i(x(k))) > 0$, and *Li–Yorke* if it is proximal and separated. A set $Y \subseteq X$ is *n -scrambled* if every injective sequence in Y^n is Li–Yorke. We say that Y is *$(<\mathbb{N})$ -scrambled* if it is n -scrambled for all $n \geq 2$.

In [GGM21], it was shown that 2-scrambled uncountable sets yield 2-scrambled Cantor sets in Polish dynamical systems. Here we show:

Theorem 1 (ZFC). *Suppose that $n \geq 2$. Then there is a homeomorphism $T: 2^{\mathbb{N}} \rightarrow 2^{\mathbb{N}}$ with an n -scrambled Cantor set containing a $(<\mathbb{N})$ -scrambled uncountable set, but with no $(n + 1)$ -scrambled Cantor set.*

Let INJ_X^n denote the set of injective sequences of elements of X of length n . Let LY_f denote the set of Li–Yorke finite sequences. A *graph* on X is an irreflexive symmetric set $G \subseteq X \times X$. We say that Y is a *G -clique* if its distinct points are G -related, and *G -independent* if its points are not G -related. Let $\text{CLQ}_G^{\geq n}$ denote the set of injective finite sequences of length at least n whose images are G -cliques, and let IND_G^n denote the set of injective sequences of length n whose images are G -independent. A *homomorphism* from a relation R on X to a relation S on Y is a function $\pi: X \rightarrow Y$ whose compositions with sequences in R are in S . Such a function is a *homomorphism* from (R, R') to (S, S') if it is also a homomorphism from R' to S' . The *saturation* of Y under a bijection $T: X \rightarrow X$ is given by $[Y]_T = \bigcup_{i \in \mathbb{Z}} T^i(Y)$. A subset of a topological space is F_σ if it is a countable union of closed sets. The primary new technical result underlying our proof of Theorem 1 is:

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Theorem 2 (ZF + DC). *Suppose that G is an F_σ graph on $2^\mathbb{N}$ and $n \geq 2$. Then there exist a homeomorphism $T: 2^\mathbb{N} \rightarrow 2^\mathbb{N}$ and a continuous injective homomorphism $\pi: 2^\mathbb{N} \rightarrow 2^\mathbb{N}$ from $(\text{INJ}_{2^\mathbb{N}}^n \cup \text{CLQ}_G^{\geq n+1}, \text{IND}_G^{n+1})$ to $(\text{LY}_T, \sim \text{LY}_T)$ for which $|\sim[\pi(2^\mathbb{N})]_T| = 1$.*

In §1, we establish Theorem 2. In §2, we obtain Theorem 1 and a related independence phenomenon as corollaries.

1. EMBEDDING HYPERGRAPHS

Define $S: 2^\mathbb{Z} \rightarrow 2^\mathbb{Z}$ by $S(c)(i) = c(i+1)$ for all $i \in \mathbb{Z}$ and $c \in 2^\mathbb{Z}$. Let SEP_f denote the set of separated finite sequences. Theorem 2 follows from the usual topological characterization of $2^\mathbb{N}$ and:

Theorem 1.1 (ZF + DC). *Suppose that G is an F_σ graph on $2^\mathbb{N}$ and $n \geq 2$. Then there exist an S -invariant perfect set $F \subseteq 2^\mathbb{Z}$, whose finite sequences are proximal, and a continuous injective homomorphism $\pi: 2^\mathbb{N} \rightarrow F$ from $(\text{INJ}_{2^\mathbb{N}}^n \cup \text{CLQ}_G^{\geq n+1}, \text{IND}_G^{n+1})$ to $(\text{SEP}_S, \sim \text{SEP}_S)$ for which $0^\mathbb{Z}$ is the unique element of $F \setminus [\pi(2^\mathbb{N})]_S$.*

Proof. For all $s \in 2^{<\mathbb{N}}$, set $\mathcal{N}_s = \{c \in 2^\mathbb{N} \mid \forall i < |s| \ s(i) = c(i)\}$. For all $S \subseteq 2^{<\mathbb{N}}$, set $\mathcal{N}_S = \bigcup_{s \in S} \mathcal{N}_s$. For all $\ell \in \mathbb{N}$, $n' \geq n$, and $p: 2^\ell \rightarrow n'$, set $\mathcal{N}_p = \prod_{i < n'} \mathcal{N}_{p^{-1}(\{i\})}$. We say that p is a *near bijection* if it induces a bijection of $2^\ell \setminus p^{-1}(\{0\})$ with $n' \setminus \{0\}$ but $\{0^\ell\} \subsetneq p^{-1}(\{0\})$. For all $0 < k < n'$, let $p^{-1}(k)$ denote the unique element of $p^{-1}(\{k\})$.

Fix an increasing sequence $(G_m)_{m \in \mathbb{N}}$ of closed graphs on $2^\mathbb{N}$ whose union is G . Define $G_{\ell,m} = \{(c, c') \in 2^\mathbb{N} \times 2^\mathbb{N} \mid G_m \cap (\mathcal{N}_{c \upharpoonright \ell} \times \mathcal{N}_{c' \upharpoonright \ell}) \neq \emptyset\}$. Let P_ℓ denote the set of all near bijections $p: 2^\ell \rightarrow n$. Let $P_{\ell,m,n'}$ denote the set of near bijections $p: 2^\ell \rightarrow n'$ such that $\mathcal{N}_{p^{-1}(j)} \times \mathcal{N}_{p^{-1}(k)} \subseteq G_{\ell,m}$ for all $0 < j < k < n'$. Fix $I \subseteq \mathbb{N} \setminus \{0\}$ and a bijection $\phi: I \rightarrow \mathbb{N} \cup \bigcup_{\ell \in \mathbb{N}} (\{\ell\} \times P_\ell) \cup \bigcup_{\ell, m \in \mathbb{N}, n' > n} (\{\ell\} \times \{m\} \times \{n'\} \times P_{\ell,m,n'})$ such that $I \cap [i, 2i + w(i)] = \{i\}$ for all $i \in I$, where $w(i) = 0$ if $\phi(i) \in \mathbb{N}$, $w(i) = n - 2$ if $\phi(i) \in \{\ell\} \times P_\ell$ for some $\ell \in \mathbb{N}$, and $w(i) = (m+1)(n' - 2)$ if $\phi(i) \in \{\ell\} \times \{m\} \times \{n'\} \times P_{\ell,m,n'}$ for some $\ell, m \in \mathbb{N}$ and $n' > n$. Define a continuous function $\pi: 2^\mathbb{N} \rightarrow 2^\mathbb{Z}$ by $\pi(c)(i) = 1$ if and only if

- (1) $i = 0$;
- (2) $i \in I$, $\phi(i) \in \mathbb{N}$, and $c(\phi(i)) = 1$;
- (3) $\exists 0 < k < n \exists \ell \in \mathbb{N} \exists p \in P_\ell$
 $((\ell, p) = \phi(i - (k - 1)) \text{ and } p(c \upharpoonright \ell) = k)$; or
- (4) $\exists n' > n \exists 0 < k < n' \exists \ell, m \in \mathbb{N} \exists p \in P_{\ell,m,n'}$
 $((\ell, m, n', p) = \phi(i - (m + 1)(k - 1)) \text{ and } p(c \upharpoonright \ell) = k)$.

Lemma 1.2. π is injective.

Proof. Suppose that $c, c' \in 2^{\mathbb{N}}$ are distinct. Then there exists $k \in \mathbb{N}$ with $c(k) \neq c'(k)$. Fix $i \in I$ for which $\phi(i) = k$. As $\pi(c)(i) = c(k)$ and $\pi(c')(i) = c'(k)$, it follows that $\pi(c)(i) \neq \pi(c')(i)$, so $\pi(c) \neq \pi(c')$. $\square$

The *product metric* on $2^{\mathbb{Z}}$ is given by $\rho(c, c') = \sum_{i \in \mathbb{Z}} |c(i) - c'(i)|/2^{|i|}$ for all $c, c' \in 2^{\mathbb{Z}}$. Define $N_i = [i, i + w(i)]$ for all $i \in I$.

Lemma 1.3. π is a homomorphism from $\text{CLQ}_G^{\geq n+1} \cup \text{INJ}_{2^{\mathbb{N}}}^n$ to SEP_S .

Proof. Suppose that $c \in (2^{\mathbb{N}})^{n'}$ is in $\text{CLQ}_G^{\geq n+1} \cup \text{INJ}_{2^{\mathbb{N}}}^n$. By permuting the coordinates of c , we can assume that $c(0) = 0^{\mathbb{N}}$ if $0^{\mathbb{N}} \in c(n')$. Suppose that $\ell \in \mathbb{N}$ is sufficiently large that $\text{proj}_{2^\ell} \circ c$ is injective, $c(k) \upharpoonright \ell \neq 0^\ell$ for all $0 < k < n'$, and $n' < 2^\ell$.

If $n' = n$, then let p be the near bijection extending $(\text{proj}_{2^\ell} \circ c)^{-1}$ in P_ℓ . Fix $i \in I$ such that $\phi(i) = (\ell, p)$. Then $\text{proj}_{2^{N_i}} \circ \pi \circ c$ is injective, so $\min_{j < k < n} \rho(S^i(\pi(c(j))), S^i(\pi(c(k)))) \geq 1/2^{n-2}$.

If $n' > n$, then fix $m \in \mathbb{N}$ such that $c(j) G_m c(k)$ for all $j < k < n'$, and let p be the near bijection extending $(\text{proj}_{2^\ell} \circ c)^{-1}$ in $P_{\ell, m, n'}$. Fix $i \in I$ such that $\phi(i) = (\ell, m, n', p)$. Then $\text{proj}_{2^{N_i}} \circ \pi \circ c$ is injective, so $\min_{j < k < n'} \rho(S^i(\pi(c(j))), S^i(\pi(c(k)))) \geq 1/2^{(m+1)(n'-2)}$. $\square$

Lemma 1.4. π is a homomorphism from IND_G^{n+1} to $\sim\text{SEP}_S$.

Proof. Suppose that $c \in (2^{\mathbb{N}})^{n+1}$ and $\pi \circ c$ is separated. Fix $\epsilon > 0$ such that $I' = \{i' \in \mathbb{N} \mid \min_{j < k < n+1} \rho(S^{i'}(\pi(c(j))), S^{i'}(\pi(c(k)))) \geq \epsilon\}$ is infinite, as well as $r \in \mathbb{N}$ such that $\sum_{|i| > r} 1/2^{|i|} < \epsilon$. By throwing out finitely many elements of I' , we can assume that for all $i' \in I'$, there is at most one $i \in I$ such that $[i' - r, i' + r] \cap N_i \neq \emptyset$. Our choice of r then yields a unique such $i \in I$, in which case $\text{proj}_{2^{N_i}} \circ \pi \circ c$ is injective. It follows that $\phi(i)$ is of the form (ℓ, m, n', p) , where $(m+1)(n-1) \leq 2r$, and there exist $j < k < n+1$ such that $c(j) G_{\ell, m} c(k)$.

By passing to an infinite subset of I' , we can assume that the values of j , k , and m are independent of $i' \in I'$. Then there are unboundedly many values of ℓ corresponding to the remaining $i' \in I'$, since otherwise there are only finitely many possibilities for both n' and p , and therefore only finitely many possibilities for i , despite the fact that the distance between any two elements of I' corresponding to the same value of i is at most $2r$. Hence $c(j) G_m c(k)$, so $c(j) G c(k)$, thus $c \notin \text{IND}_G^{n+1}$. $\square$

The closure F of the S -saturation of $\pi(2^{\mathbb{N}})$ is clearly perfect.

Lemma 1.5. The sequence $0^{\mathbb{Z}}$ is the unique element of $F \setminus [\pi(2^{\mathbb{N}})]_S$.

Proof. If $(c_k)_{k \in \mathbb{N}}$ is a sequence of elements of $2^{\mathbb{N}}$ and $(i_k)_{k \in \mathbb{N}}$ is a strictly increasing sequence of elements of $\mathbb{N}$, then $S^{-i_k}(\pi(c_k)) \rightarrow 0^{\mathbb{Z}}$. And if

$S^{i_k}(\pi(c_k))$ converges to a sequence $c \in 2^{\mathbb{Z}} \setminus \{0^{\mathbb{Z}}\}$, then $c(i) = 1$ for a unique $i \in \mathbb{Z}$, so it is in the S -orbit of $\pi(0^{\mathbb{N}})$. $\square$

Lemma 1.5 implies that the union of the supports of finitely many elements of F is contained in a finite union of translates of $\{0\} \cup \bigcup_{i \in I} N_i$. But such sets necessarily have arbitrarily long gaps, so every finite sequence of elements of F is proximal. $\square$

Proposition 1.6 (ZF + DC). *Suppose that $n \geq 3$, X and Y are Polish spaces, G is a Borel graph on X , $T: Y \rightarrow Y$ is a Borel automorphism, $B \subseteq Y$ is an n -scrambled uncountable Borel set, $\pi: X \rightarrow Y$ is an injective Borel homomorphism from IND_G^n to $\sim\text{LY}_T$, and $[\pi(X)]_T$ is co-countable. Then G has a perfect clique.*

Proof. Fix $i \in \mathbb{Z}$ for which $B \cap (T^i \circ \pi)(X)$ is uncountable. The perfect set theorem and Galvin's Theorem (see, for example, [Kec95, Theorems 13.6 and 19.7]) then yield a perfect set $P \subseteq (T^i \circ \pi)^{-1}(B)$ that is either a G -clique or G -independent. But G -independence would contradict the fact that B is n -scrambled. $\square$

2. n -SCRAMBLED SETS

We begin this section with the proof of our primary result:

Proof (of Theorem 1). By [KV12, Theorem 2.1] and a straightforward Baire category argument, there is an F_σ graph G on $2^{\mathbb{N}}$ with an uncountable clique $Y \subseteq 2^{\mathbb{N}}$ but no perfect clique. By Theorem 2, there exist a homeomorphism $T: 2^{\mathbb{N}} \rightarrow 2^{\mathbb{N}}$ and a continuous injective homomorphism $\pi: 2^{\mathbb{N}} \rightarrow 2^{\mathbb{N}}$ from $(\text{INJ}_{2^{\mathbb{N}}}^n \cup \text{CLQ}_G^{\geq n+1}, \text{IND}_G^{n+1})$ to $(\text{LY}_T, \sim\text{LY}_T)$ for which $|\sim[\pi(2^{\mathbb{N}})]_T| = 1$. Then $\pi(2^{\mathbb{N}})$ is an n -scrambled Cantor set, $\pi(Y)$ is a $(<\mathbb{N})$ -scrambled uncountable set, and Proposition 1.6 rules out the existence of an $(n+1)$ -scrambled Cantor set. $\square$

Let $\mathfrak{c}$ denote the cardinality of the continuum.

Theorem 2.1 (ZFC + $\neg\text{CH}$). *It is consistent that, for all $n \geq 2$, there is a homeomorphism $T: 2^{\mathbb{N}} \rightarrow 2^{\mathbb{N}}$ that has an n -scrambled Cantor set containing a $(<\mathbb{N})$ -scrambled set of cardinality $\mathfrak{c}$, but does not have an $(n+1)$ -scrambled Cantor set.*

Proof. By [She99, Theorem 1.13], we can assume that there is an F_σ graph G on $2^{\mathbb{N}}$ with a clique of cardinality $\mathfrak{c}$ but no perfect clique. The proof of Theorem 1 therefore yields the desired result. $\square$

A *hypergraph* is a permutation-invariant set of injective sequences.

Theorem 2.2 (ZFC + $\neg$ CH). *It is consistent that, for all $n \geq 2$, every Borel function on a Polish space with an n -scrambled set of cardinality $\aleph_2$ has an n -scrambled perfect set.*

Proof. By [KS03, Proposition 3.4], we can assume that every Borel n -ary hypergraph on a Polish space with a clique of cardinality $\aleph_2$ has a perfect clique. But n -ary Li–Yorke-ness is a Borel hypergraph. $\square$

Remark. The proof of [KS03, Proposition 3.4] can be used to establish the analog of Theorem 2.2 for $(<\aleph_2)$ -scrambled sets.

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